

SYED AMRULLAH

✉ vtu26294@veltech.edu.in ☎ 9087850676 📍 Chennai 🔗 LinkedIn 🐙 GitHub

PROFESSIONAL SUMMARY

Results-driven Computer Science and Engineering student with hands-on experience in Artificial Intelligence, Machine Learning, Computer Vision, IoT, Embedded Systems, and Full Stack Development. Proficient in Python, TensorFlow, YOLOv8, OpenCV, ONNX Runtime, JavaScript, REST APIs, and Raspberry Pi 5. Built AI-powered solutions for industrial safety, healthcare monitoring, and smart surveillance using deep learning and Edge AI technologies. Internship experience in Frontend Development with knowledge of Agile methodology, SDLC, model optimization, and deployment workflows. Seeking opportunities in AI/ML Engineering, Software Development, or IoT Systems.

PROFESSIONAL EXPERIENCE

Frontend Development Intern

06/2025 – 09/2025 | Remote, Chennai

Greensoft Groups [🔗](#)

- Developed responsive web interfaces using HTML5, CSS3, and JavaScript following modern frontend development practices.
- Worked on UI design, webpage structuring, and responsive layouts to improve usability across multiple devices.
- Collaborated in a team-based development environment while gaining exposure to Agile methodology and SDLC practices.
- Assisted in debugging, testing, and optimizing frontend components for performance and user experience.

EDUCATION

B.Tech Computer Science and Design

2023 – 2027 | Chennai, India

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

CGPA- 8.70

SKILLS

Programming Languages: Python, Java, C, C++, JavaScript, SQL

AI/ML: Machine Learning, Deep Learning, Computer Vision, CNN, NLP, LLMs, Generative AI, Prompt Engineering, Object Detection, Image Classification, Edge AI, Transfer Learning, Predictive Analytics

Frameworks & Libraries: TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv8, ONNX Runtime, Flask, Pandas, NumPy, Matplotlib

Web Development: HTML5, CSS3, JavaScript, REST APIs, Responsive Web Design, Frontend Development

IoT & Embedded Systems: Raspberry Pi 5, ESP32, IoT Sensor Integration, Embedded Systems, Edge Device Deployment, I2C Communication

Tools & Platforms: Git, GitHub, Linux, VS Code, Jupyter Notebook, Kaggle, Ollama

PROJECTS

AI-Based Industrial PPE Detection System [🔗](#)

- Engineered a real-time PPE compliance detection system using YOLOv8 and ONNX-based helmet classification on Raspberry Pi 5.
- Optimized CPU-only edge inference using ONNX Runtime and multithreaded OpenCV for low-latency performance.
- Integrated ROI-based detection and automated SMTP alert monitoring for industrial safety compliance.
- Configured Linux deployment with VNC monitoring and system auto-startup for continuous operation.

Livestock Water Monitoring & Microbial Risk Prediction System

- Built an IoT-based livestock water quality monitoring system integrating temperature, turbidity, and water-level sensors.
- Enabled real-time multi-parameter sensor data acquisition using I2C and 1-Wire communication protocols.
- Created a microbial risk prediction framework for proactive livestock health monitoring.
- Architected a scalable system for predictive analytics, anomaly detection, and automated alert generation.

Women Safety Analytics – Real-Time Threat Detection System

- Developed an AI-powered surveillance system for real-time women safety threat detection using computer vision techniques.
- Implemented person detection, gender classification, and crowd density analysis for high-risk scenario identification.
- Designed ML-based and rule-based alert mechanisms for suspicious activity detection.
- Enhanced automated monitoring workflows for public safety analytics applications.

Smart Wearable for Oxygen Desaturation Monitoring in ILD Patients

- Designed a low-power wearable device for continuous real-time SpO2 monitoring in ILD patients.
- Proposed a VAE-GAN-based anomaly detection model for early desaturation event prediction.
- Optimized lightweight AI inference for ESP32 edge deployment and wearable computing applications.

CERTIFICATES & ACHIEVEMENTS

VISAI Technical Certification [🔗](#)

Cloud Computing Workshop Certification [🔗](#)

Innovers Expo Technical Contest Participant [🔗](#)

Hacktrix 2025 Participation Certificate [🔗](#)